

PROBLEM SET 6

ECO403/503–INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Instructions: Due on 28/11/2021; scan or type your work and submit through the link in Moodle. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission.

1. Consider a principal who hires a salesman to approach a client. The reservation utility level of the salesman is 3. A high level of effort gives no sale with probability 0.2, a \$100 sale with probability 0.4 and a \$400 sale with probability 0.4. For a low level of effort, the three probabilities are 0.4, 0.4, 0.2 respectively.

The principal is risk neutral. The agent is risk averse, and if the wage is w and the effort is a , he gets utility $u(w, a) = \sqrt{w} - a$. (High effort requires $a = 3$ and low effort requires $a = 0$.)

- (a) For starters, assume that the principal could observe the level of effort put by the salesman. Characterize the first-best wage schedule.
 - (b) From now on, assume that contracts can be made contingent on the size of the sale only (effort is unobservable). What is the optimal way to induce the salesman to put in a low level of effort?
 - (c) What is the optimal way to induce the salesman to put in a high level of effort?
 - (d) What is the optimal contract (from the principal's perspective) to offer the salesman?
2. Consider the following moral hazard insurance model. Let the consumer's utility function for wealth be $u(w) = \sqrt{w}$, his initial wealth be $w_0 = \$100$, and suppose that there are two loss levels, $l = 0$ and $l = \$51$. As usual, there are two effort levels $e = 0$ and $e = 1$. The consumer's disutility of effort is given by the function $d(e)$, where $d(0) = 0$ and $d(1) = 1/3$. Finally, suppose that the loss probabilities are given by the following table, in which rows correspond to effort and column to loss levels:

	$l = 0$	$l = 51$
$e = 0$	1/3	2/3
$e = 1$	2/3	1/3

- (a) Verify that the probabilities given in the table satisfy the monotone likelihood ratio property.
- (b) Find the consumer's reservation utility assuming that there is only one insurance company and that the consumer's only other option is to self-insure. *Hint: this exercise provides a justification of where the reservation utility comes from. You need to compute what level of utility the consumer can achieve without insurance.*
- (c) What effort level will the consumer exert if no insurance is available? *Hint: This continues the previous part. Without insurance, the consumer will exert the effort level that achieves his reservation utility.*
- (d) Show that if information is symmetric (first-best), then it is optimal for the insurance company to offer a policy that induces high effort.
- (e) Show that the policy in part (d) will not induce high effort if information is asymmetric.
- (f) Find the optimal policy when information is asymmetric.
- (g) Compare the insurance company's profits in the symmetric and asymmetric information cases. Also, compare the consumer's utility in the two cases. Argue that the symmetric information solution Pareto dominates (is more efficient) than with asymmetric information.